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A longitudinal analysis of the growth of limb segments in adolescence

N. CAMERON, J. M. TANNER and R. H. WHITEHOUSE

Department of Growth and Development, Institute of Child Health, University of London

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Summary. The growth of upper and lower-limb segments of 96 adolescent boys and girls from the Royal Hospital School Longitudinal Study was analysed. Preece–Baines Model 1 curves were fitted to the longitudinal data to obtain, for each measurement, age at peak velocity and the magnitude of this velocity.

Mean-constant peak velocities were between 1·0 and 2·5 cm/yr in all segments. They were in all cases greater than the values obtained from fitting the P–B curve to the cross-sectional means at successive ages. Boys had greater peak velocities than girls in all measurements (sex ratio 1·1 to 1·4). On average distal segments preceded more proximal segments in the ages at which peak velocity occurred. Considerable individual differences, however, occurred in the order for the upper limb segments. These differences seemed to be related to the individual's tempo of growth; late developers had a significantly different order to early developers.

1. Introduction

Investigations of the growth of the limbs are few in number and mostly cross-sectional. There is some doubt about whether all segments have pubertal growth spurts, and about the order in which the spurts occur. Tanner (1962) supposed that there were proximal–distal growth maturity gradients in both limbs in man, but based his analysis only on plotting percentage of adult measurement reached at successive ages. There is no necessarily exact relationship between this method for demonstrating gradients and the method of considering ages for peak velocities of the measurements in question as done by Welon and Bielicki (1979). These authors analysed longitudinal data on 192 boys and 250 girls to investigate the relationship of eight body dimensions which, in the limbs, included hand and foot lengths only. They demonstrated clear growth spurts in these dimensions, but the absence of longitudinal analyses of the growth of the other limb segments has prevented a consistent picture of the relationships between body dimensions emerging.

Other studies of the upper-limb in the live subject have been based on radiographic, photogrammetric and anthropometric methods. Some radiographic studies have measured the long bones excluding their epiphyses (Gindhart 1973, Maresh 1943); others have included them (Maresh 1943, 1955). Photogrammetric studies have been made on young adults (Harrison and Marshall 1970, Marshall and Harrison 1971) and children (Marshall and Ahmed 1976, Ahmed 1976, Attalah 1980). Anthropometric results are available from the Brush Foundation Study (Simmons 1944), the Philadelphia Growth Study of black and white children (Krogman 1970), the Iowa Child Welfare Research Station Study by Wallis (1931) and from Chinese children (Low 1978, Chang, Chen and Low 1968) and Czech children (Flatt and Burmeister 1979).

Literature on the lower limb is scantier. Early work was by Davenport (1932, 1944); then followed a series of studies by Andersen and his colleagues (Andersen, Blaise and

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Green 1956, Andersen, Green and Messner 1963, Andersen, Messner and Green 1964) and most recently by Meredith (1978).

The purpose of the present investigation was (1) to provide longitudinal data on the timing and magnitude of adolescent growth spurts in limb segments, the sex ratios of these spurts and their interrelationship with spurts in stature, sitting height and subischial leg length, (2) to elucidate the order of occurrence of these spurts in boys and girls and (3) to compare 'percentage of adult measurement' and 'age at peak velocity of a measurement' as maturity indicators.

2. Subjects and methods

The subjects were from the Royal Hospital School Longitudinal Growth Study (Clark and Zorab 1978). All were pupils at boarding schools where they resided continuously during the school term. They were in good physical health at the time of the examination; any child who was temporarily ill at the scheduled time of examination was excluded on that occasion.

Examinations took place at the beginning of each school term, i.e. January, May and October. Boys entered the study on first arrival at the school, at which time they were aged between 11 and 12 years. Most girls entered between the ages of 9·5 and 10·5 years; a few were younger or older. The data on 52 boys and 44 girls, with ages ranging from 11·1 to 18·0 years for boys and 8·8 to 16·5 years for girls, were analysed. Two of the boys and ten of the girls had missing examinations. The minimum number of examinations per child was nine, but the majority of children had between 16 and 19 examinations, the most frequent number being 16.

Anthropometric measurements

Sixteen anthropometric measurements were taken on each subject, but only height (HT), sitting height (SH), subischial leg length (SLL) and lengths of the upper arm (UA), forearm (FA), hand (HD), tibia (CF) and foot (FT) were used in this analysis. All subjects were measured on every occasion by R. H. Whitehouse. Height and sitting height were taken as described by Tanner (1962) and the lengths of the limbs as described by Cameron (1978). The Harpenden Stadiometer was used to measure height and the Harpenden Anthropometer for all limb measurements. Both instruments were calibrated before each measurement session. Measurements were recorded by the same trained recorder on all occasions and immediately compared with previous values. If any measurement appeared dubious, it was repeated at the end of the session and then repeated again if a difference of 5 mm occurred. No further steps beyond those already described were taken to edit or clean the data.

Longitudinal analysis

The logistic and Preece–Baines Model 1 (Preece and Baines 1978) curves were fitted to the data of each individual. Preece–Baines Model 1 (PB1) gave lower residual mean squares when pooled over all subjects by sex, although some individuals did have lower mean squares for some variables on the logistic fit. To provide consistency in analysis, PB1 was used throughout. The data for some individuals could not be fitted by the curve. These were either subjects who had already begun their adolescent growth spurts and were at, or past, peak velocity or subjects who by the end of the study were only just entering their growth spurts. Thus the sample size is slightly different for each anthropometric measurement, although a central core group of 29 boys and 20 girls were fitted on all variables. Table 1 lists the sub-samples for each variable.

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Table 1. Sub-sample sizes for each anthropometric variable.

Dimensions	Boys *N*	Boys % of total (52)	Girls *N*	Girls % of total (44)
All dimensions	29	56	20	46
Upper limb	35	67	27	61
Lower limb	41	79	31	70
HT, SH, SLL (total body)	49	94	33	77
Total body and upper limb	35	67	26	59
Total body and lower limb	40	77	27	61

The timing and magnitude of growth spurts were examined by comparisons of the biological parameters derived from PB1, e.g. age at peak velocity, and amount of peak velocity. Sex ratios and interrelationships between limb segments and height, sitting height and subischial leg length were examined by comparing peak-velocity ages of each variable.

3. Results

Table 2 presents the mean-constant values obtained by fitting PB1 to the maximum number of subjects who could be fitted in all segments of the upper limb and all segments of the lower limb. Peak velocities of limb segments in both sexes are all around 1 to 2 cm/yr and their average times of occurrence fall in boys between 13·5 and 14·2 years, depending on the segment, and in girls between 12·2 and 12·8 years. The values in brackets in table 2 are those obtained by fitting PB1 to the cross-sectional means (values for boys' hand and girls' tibia could not be reasonably determined by smoothing the cross-sectional means). The cross-sectional values at peak velocity are in all cases lower than the mean-constant, longitudinal ones. The age of occurrence of

Table 2. Mean-constant values at peak velocity (PV). The figures in brackets are from the cross-sectional means fitted by PB1.

Lengths	*N*	PV age (yrs)	PV distance (cm)	PV value (cm/yrs)
Boys				
Upper arm	35	13·98	29·79	1·77
		(12·03)	(27·02)	(1·22)
Forearm	35	14·16	23·78	1·76
		(14·58)	(24·23)	(1·12)
Hand	35	13·49	17·15	1·34
Tibia	41	14·13	37·36	2·48
		(12·83)	(35·41)	(1·99)
Foot	41	13·81	24·83	1·41
		(13·30)	(24·55)	(1·04)
Girls				
Upper arm	27	12·62	28·71	1·55
		(11·95)	(27·95)	(1·22)
Forearm	27	12·75	22·42	1·33
		(12·45)	(22·14)	(0·84)
Hand	27	12·37	16·34	1·07
		(10·59)	(15·25)	(0·64)
Tibia	31	12·62	36·16	2·16
Foot	31	12·18	22·79	1·03
		(11·05)	(22·15)	(0·71)

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peak velocity was mostly lower in cross-sectional than in longitudinal fits, on average by about 0·8 years.

Boys had greater magnitudes of peak velocities in all dimensions. The sex ratios (boys/girls) ranged from 1·14 for the upper-arm to 1·37 for the foot. Height, sitting height and sub-ischial leg length had sex ratios of 1·28, 1·26 and 1·11 respectively.

The central-core group of 29 boys and 20 girls was used to investigate the order of occurrence of peak velocities. Table 3 presents the peak velocity ages for each dimension (1) when the maximum number of subjects were fitted and (2) when only the core-groups for each sex were fitted. The order in which peak velocity occurs is similar in both sexes in both the total and core groups—Spearman's rank order correlations being 0·83 and 0·62 respectively. The orders of the total groups and core-groups within sex are also similar, rank order correlations being 0·86 and 0·81 for boys and girls. Thus the order in which the limb segments exhibit their peaks is similar in boys and girls and is not obscured by using the smaller core groups.

Figure 1 illustrates the mean ages of peak velocity (± 2 SE) in both sexes. Body length segments appear to follow the established pattern of subischial leg length spurting earlier than sitting height. Boys show a smaller mean time difference between subischial leg length and sitting height than girls (0·3 years and 0·6 years respectively). The girls have greater SEs, which may reflect a lack of homogeneity in this particular sample.

When the mean peak velocity ages of limb *and* body segments are investigated, the order in boys is HD, FT, SLL, CF, HT, UA, SH, FA and in girls FT, SLL, HD, HT, FA, CF, SH, UA. This order is statistically similar in both sexes (Spearman's $r = 0{\cdot}71$).

The mean time taken to pass through this sequence of events is the same in boys and girls, 0·61 years. The peak velocities, of course, do not occur at fixed time intervals within the 0·61 years. Figure 2 illustrates the temporal relationship between the peaks. Once again the girls are far more variable than boys in mean time differences.

It is perhaps misleading to assume that the order of events in a given individual is the same as that shown by the group means. Inspection of the 29 boys and 20 girls, individually, demonstrated considerable departures from the group order (table 4).

Table 3. Peak velocity ages when (1) the maximum number of subjects were fitted and (2) when only the central core group was fitted by PB1 curve.

	Boys		Girls	
Length	*N*	PV age (yrs)	*N*	PV age (yrs)
Stature	(1) 50	13·90	42	12·16
	(2) 29	14·10	20	12·59
Sitting height	(1) 51	14·12	38	12·72
	(2) 29	14·25	20	12·85
Subischial leg length	(1) 51	13·88	41	12·14
	(2) 29	13·98	20	12·26
Upper arm	(1) 42	14·02	36	12·41
	(2) 29	14·23	20	12·86
Forearm	(1) 45	14·21	35	12·69
	(2) 29	14·29	20	12·65
Hand	(1) 49	13·56	35	12·25
	(2) 29	13·68	20	12·37
Tibia	(1) 42	14·08	36	12·67
	(2) 29	13·97	20	12·74
Foot	(1) 50	13·67	38	12·13
	(2) 29	13·76	20	12·25

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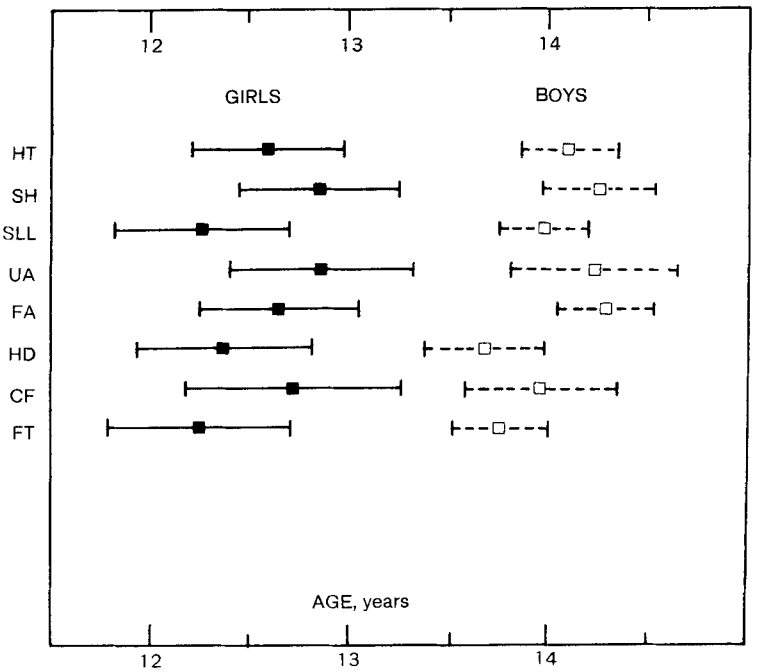

Figure 1. Mean peak velocity age (±2 SE) of the central core groups.

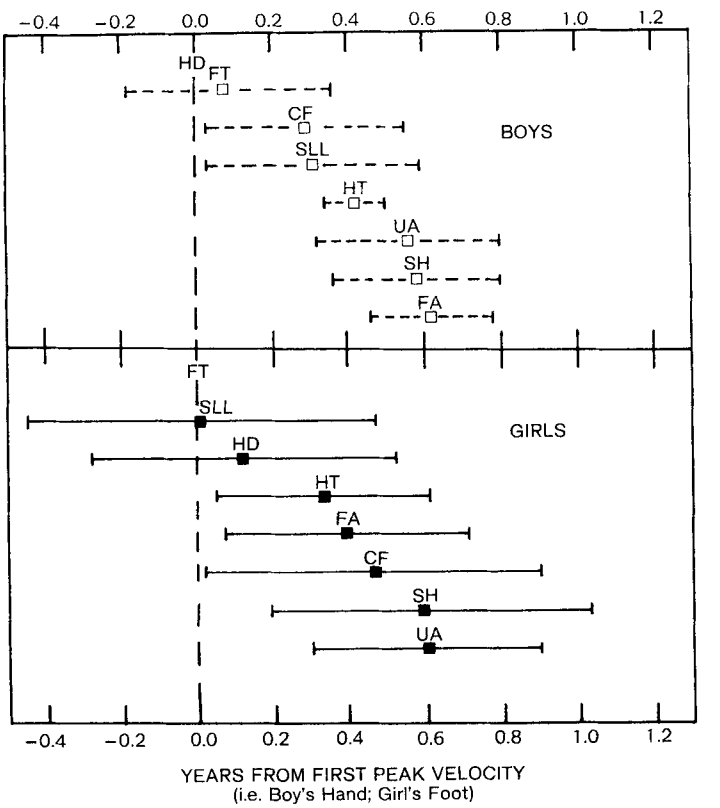

Figure 2. Mean time difference between peak velocity ages in boys and girls.

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Sixty-five per cent of both boys and girls followed the accepted body pattern of SLL-HT-SH, but 15% of boys and 27% of girls had SH peak before SLL peak. In the upper limb 41% of boys and 35% of girls showed peak velocity ages in the order HD-FA-UA. In the lower-limb foot length, peak velocity preceded calf peak velocity in 69% of boys and 75% of girls. A similar pattern of hand preceding forearm occurred in 86% of boys and 65% of girls. It may be concluded therefore that while considerable numbers of individuals follow the accepted distal–proximal gradient of FT, CF in the lower limb and HD, FA, UA in the upper limb, this is not true of all children. Indeed, individuals may be found who demonstrate all possible orders of peak velocity ages.

The order of events in the upper limb was further investigated by analysing the samples forming the upper-limb core group. These were children who were fitted to all upper limb segments regardless of whether they were also fitted to the body and lower limb segments. The samples are therefore larger than for the central core group. The most frequent pattern (12 boys and 7 girls) was HD, FA, UA as in the core group, but the pattern of HD, UA, FA was shown by almost as many individuals (10 boys and 4 girls) as well as by the means. A systematic difference in tempo of growth is evident, however, between these two sets of children. The HD, FA, UA children had mean PHV ages of 14·6 (boys) and 12·8 years (girls)—that is, they were late developers—while the HD, UA, FA children had PHV ages of 13·4 and 11·8 years—they were early developers. The difference of about a year in PHV is statistically significant in boys ($t=5{\cdot}0$; $P=0{\cdot}001$), but fails to reach the 5% level in girls ($t=2{\cdot}1$; $P=0{\cdot}11$).

Graphs of percentage of adult value plotted against chronological age are shown in figure 3. PB1 parameter 1 was used as the adult (mature) value; since this is an asymptotic value it is probably a better estimate of adult values than the oldest measured value. In boys at least the distal–proximal pattern of HD, FA, UA is extremely clear; hand values are nearer the adult value at any age than forearm or upper arm. In girls, however, the pattern is less clear. The upper arm is certainly less

Table 4. Percentage of sample exhibiting particular orders of peak velocity in body segments and the upper and lower limb

Order of attaining peak velocity	Boys ($N=29$)		95% Confidence limits	Girls ($N=20$)		95% Confidence limits
	N	%		N	%	
Body segments						
SLL–HT–SH	19	65·5	46–82	13	65·0	41–85
SH–HT–SLL	4	13·8	4–32	2	10·0	1·–32
HT–SLL–SH	1	3·4	0–12	3	15·0	3–38
HT–SH–SLL	4	13·8	4–32	1	5·0	0–17
SLL–SH–HT	—	—		1	3·4	0–17
SH–SLL–HT	1	5·0	0–12	—	—	
Upper limb						
HD–FA–UA	12	41·4	24–61	7	35·0	15–59
HD–UA–FA	6	20·7	8–40	3	15·0	3–38
UA–HD–FA	7	24·1	10–44	3	15·0	3–38
FA–HD–UA	2	6·8	1–23	3	15·0	3–38
FA–UA–HD	1	3·4	0–12	3	15·0	3–38
UA–FA–HD	1	3·4	0–12	1	5·0	0–17
Lower limb						
FT–CF	20	69·0	49–85	15	75·0	51–91
CF–FT	9	31·0	15–51	5	25·0	9–49

Abbreviations for body and limb segments are explained under *Anthropometric measurements*.

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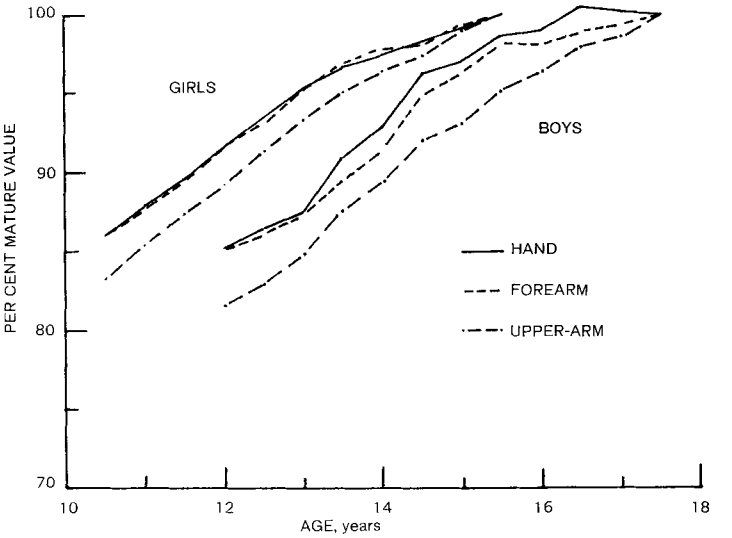

Figure 3. Percentage maturity of upper limb segments at ages from final value.

mature than either of the two other segments, but the forearm and hand do not differentiate clearly. Of the 10 plots between 10·5 and 15·5 years, the forearm has greater values than the hand in five.

Correlations between ages at peak velocity of stature, sitting height and subischial leg length, and peak velocity, size at peak, percentage adult at peak and the adult value are given in table 5, together with those reported by Tanner, Whitehouse, Marubini and Resele (1976). All correlations between peak velocity and age at peak velocity are negative (some not significantly so, but the samples are small) and there is good agreement between the two studies (if one allows for an error in Tanner *et al.* (1976) which introduced negative signs in 9 of the 10 correlations between age at peak velocity and percentage adult size at peak velocity; all correlations should be positive).

4. Discussion

Literature on the upper limbs suggests that peak velocities occur in upper-arm length at about 13·0 years in boys and 11·0 years in girls with values of 1·7 cm/yr and 1·5 cm/yr respectively (Maresh 1943, Krogman 1970). Forearm length has been reported as greater in boys at *all* ages. Peak velocities have been shown at ages varying from 12·5 to 14·0 years in boys and 10·0 to 11·5 years in girls with values ranging from 1·1 to 1·5 cm/yr and 1·05 to 1·3 cm/yr respectively (Simmons 1944, Maresh 1943, Gindhart 1973, Krogman 1970). Welon and Bielicki (1979) are unique in reporting a growth spurt in the hand at 14·0 years in boys and 11·5 years in girls at magnitudes of "about or less than 1 cm/yr". Andersen, Blaise and Green (1956) compared the growth of the foot and lower limb to stature. Peak velocities were reported for the foot at 12·5 years in boys and 10·5 years in girls at about 1·0 cm/yr. Welon and Bielicki (1979) reported similar magnitudes for peak velocity occurring at 13·0 years in boys and 10·6 years in girls. The relative lateness of peak foot in the RHS sample (13·8 boys, 12·2 girls) compared to these other two is not easily explained; it is unlikely to be due to the absence of early developers from our sample, since only two boys and six girls could not be fitted by PB1 for foot length, but may be due to differences in statistical method.

The results presented here indicate that when longitudinal data are appropriately analysed, boys have greater magnitudes of peak velocity than girls in all dimensions.

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Table 5. Correlation between age at peak velocity and magnitude at peak velocity (PV), adult value (A), size at peak velocity (size PV) and percentage adult height at peak velocity (% adult at PV).

	Height		Sitting height		Subischial leg length	
	H.L.S.†	R.H.S.‡	H.L.S.	R.H.S.	H.L.S.	R.H.S.
Age at PV versus PV						
Boys	−0·50**	−0·46*	−0·30*	−0·21	−0·42**	−0·45**
Girls	−0·29	−0·15	−0·15	−0·13	−0·13	−0·10
Age at PV versus A						
Boys	0·02	−0·10	−0·17	−0·32	0·18	0·00
Girls	−0·03	−0·17	−0·06	−0·51*	0·05	−0·03
Age at PV versus size PV						
Boys	0·27*	0·13	0·16	0·13	0·30*	0·11
Girls	0·21	0·16	0·23	0·08	0·29	0·44*
Age at PV versus % adult at PV						
Boys	0·68***	0·68***	0·60***	0·49**	0·40**	0·46**
Girls	0·57***	0·65***	0·46**	0·63**	0·63***	0·72***

Significance ***0·1%; **1·0%; *5·0%.
† Harpenden Longitudinal Study, see text.
‡ Royal Hospital School, this study.

Also peak velocities are generally higher than those previously reported after cross-sectional analysis. The ages at peak in this sample are also slightly greater than those in the literature. It is evident from the raw data that negative increments may be obtained because of soft-tissue changes occurring at the measurement points. Such increments make simple linear interpolations between data points unrepresentative of actual growth and require that serial data be smoothed to determine the real growth curve. When velocities are of small magnitude the presence of a peak during adolescence may well be lost by not analysing serial data using a curve-fitting approach, or by increasing measuring error, for example, by changing the observers. The differences in peak velocity age between longitudinal and cross-sectional mean data (table 2) presumably reflect the effects of tempo differences between individuals. Such differences may be fairly small when dealing with a variable such as height in which velocities are of relatively high magnitude, but in analyses of the low velocities common to limb segments, spurious peak velocities may result from elevated means before actual peak velocity. Interestingly, the Preece–Baines Model 1, which fitted all variables well in individuals, was unable to determine reasonable values from the cross-sectional means of the boys' hand or girls' tibia lengths. Analysis of the data longitudinally, however, allows clear peaks to be determined at appropriate ages. Andersen *et al.* (1956) suggest that boys' feet are always greater in size than girls'; the present results do not support that suggestion. The curves of raw data and the fitted curves both suggest that a cross-over occurs at about 10 years.

It is perhaps surprising that some subjects (6 boys, 4 girls) exhibited PHV before or after peaks in sitting height and SLL when, logically, PHV should occur between these two events. However, in all cases in which this occurred, the difference in age between PHV and the nearest other peak age was less than 0·07 years. There is, of course, an error associated with the determination of peak velocity ages and such small differences are likely to be within this error. Also there is the added complication that sitting height (and therefore SLL) may also be affected by changes in body fat, which were not

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analysed in this study, but may well account for small shifts in the peaks of sitting height and SLL.

The present investigation confirms that the general trend for limb maturity gradients to be distal to proximal, but considerable individual variation exists. The hands and feet tend to exhibit peak velocities earlier than the forearm and tibia, but the humerus (and presumably the femur) may exhibit growth spurts between these two dimensions. 'Growth gradients' have previously been reported based on the interpretation of distance graphs of percentage adult value against age (Tanner 1962; p. 76). Using this analytical technique a clear pattern of distal to proximal is evident in the boys' graphs, but the hand shows no clear dominance over the forearm in girls.

Welon and Bielicki (1979) found no differences in the order in which peak velocities occurred in early and late developers, in contrast to our clear difference for the upper limb segments in both sexes. Such differences between comparable studies are difficult to explain; it requires larger samples of early and late developers to clarify whether or not they do, in fact, exist.

It is evident that 'age at peak velocity' and 'percentage adult' are different indicators of maturity. Both may be used as instantaneous values inferring a particular tempo of growth and at such times are significantly correlated (see table 5). Yet the latter may be followed over time and, as it is derived from a distance value, is not so much reflecting tempo as the result of tempo. Thus a child with an early PHV will have a greater percentage mature height at any age during adolescence than one with a late PHV, since PHV does not occur at a fixed percentage of adult size. Thus the two indicators are not interchangeable.

It is apparent that the investigation of adolescent growth of the body's segments requires longitudinal analysis to elucidate both the inter- and intra-individual differences. This approach becomes obligatory when studying the presence of growth gradients and their association with an individual's tempo of growth during adolescence.

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Address correspondence to: Dr N. Cameron, Department of Growth and Development, Institute of Child Health, 30 Guilford Street, London WC1N 1EH.

Zusammenfassung. Das Wachstum von Abschnitten der oberen und unteren Gliedmaßen von 96 heranwachsenden Knaben und Mädchen aus der Längsschnittuntersuchung der Royal Hospital School wurde untersucht. Die Kurven des Preece–Baines–Modells I wurden an die Längsschnittdaten angepaßt, um für jedes Maß das Alter beim größten Zuwachs und die Größe dieses Zuwachses zu bestimmen.

Die mittleren größten Zuwächse lagen zwischen 1,0 und 2,5 cm/Jahr für alle Abschnitte. Sie waren in allen Fällen größer als die Werte, die man von der Anpassung der P-B-Kurve an die Querschnittsmittelwerte aufeinanderfolgender Altersstufen erhält. Knaben hatten bei allen Maßen höhere größte Zuwächse als Mädchen (Geschlechterverhältnis 1,1 : 1,4). Im Durchschnitt waren die distalen Abschnitte den proximaleren Abschnitten in den Altern voraus, in denen der größte Zuwachs stattfand. Beträchtliche individuelle Unterschiede kamen jedoch bei der Reihenfolge der Segmente der oberen Gliedmaßen vor. Diese Unterschiede schienen mit dem Wachstumstempo eines Individuums verknüpft; Spätentwickler hatten eine signifikant andere Reihenfolge als Frühentwickler.

Résumé. La croissance des segments des membres supérieurs et inférieurs de 96 garçons et filles adolescents de l'Etude Longitudinale de l'Ecole de l'Hôpital Royal a été analysée. Des courbes du Modèle I de Preece–Baines ont été ajustés aux données longitudinales de façon à obtenir, pour chaque mensuration, l'âge au pic de vitesse et la grandeur de cette vitesse.

Les vitesses au pic à moyenne constante étaient entre 1,0 et 2,5 cm par an pour tous les segments. Elles étaient dans tous les cas supérieures aux valeurs obtenues par ajustement de la courbe de P-B aux moyennes transversales aux âges successifs. Les garçons avaient des vitesses supérieures au pic plus grandes que celles des filles pour toutes les mensurations (sex ratio de 1,1 à 1,4). En moyenne, les segments distaux précédaient les proximaux quant à l'âge auquel le pic de vitesse survenait. Cependant, des différences individuelles considérables se marquaient dans l'ordre concernant les segments du membre supérieur. Ces différences semblaient en relation avec le tempo de croissance de l'individu; les sujets à développement tardif avaient un ordre significativement différent de celui des sujets à développement précoce.

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